ResearchShop: A Desktop Application for Auditable Gene and Variant Extraction from Biomedical Literature Using Hybrid NER–LLM Architecture

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ABSTRACT

We present ResearchShop, a free, open-source desktop application that assists biomedical researchers with structured extraction of gene and variant associations from PubMed literature. The system combines PubTator3 Named Entity Recognition, open-access full-text parsing from PubMed Central and Europe PMC, and Google Gemini 2.5 Flash Lite extraction behind a validation-oriented pipeline. Its safeguards include HGNC gene symbol verification, source-text grounding checks, HGVS-style variant pattern validation, citation cross-referencing using dense sequence matching, and explicit provenance fields for each output row. The application runs on consumer hardware with no ResearchShop server infrastructure; users provide their own Gemini API key and NCBI Entrez email for direct calls to external services.

The contribution of ResearchShop is a reproducible desktop workflow for paper discovery, open-access full-text acquisition, candidate gene discovery, LLM-assisted structured extraction, validation, and CSV/XLSX export. The project includes electron-builder packaging targets for macOS, Windows, and Linux, an offline test suite for core parser and validation behavior, and audit artifacts that help users inspect candidate provenance, gate drops, confidence scores, and extraction failures. ResearchShop is intended as a research acceleration tool; extracted associations require expert review before use in publications, systematic reviews, or clinical contexts.

Metadata

Nr.	Code metadata description	Metadata
C1	Current code version	~~v1.0.1~~v1.0.2
C2	Code/repository link	`https://github.com/michaluppal/Researchshop-Disease2Gene`
C3	Compute environment	Local desktop (Electron + Python virtual environment)
C4	Legal code license	Apache License 2.0
C5	Version control	git
C6	Software code languages, tools, and services used	TypeScript, React, Electron, Python 3.11+, BioPython, pandas, google-genai (Gemini API), electron-builder, Tailwind CSS
C7	Compilation requirements, operating environments, and dependencies	Node.js 18+, Python 3.11+; macOS, Windows, Linux
C8	Developer documentation/manual	`https://github.com/michaluppal/Researchshop-Disease2Gene/blob/main/README.md`
C9	Support email for questions	michal.uppal@gmail.com

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1. Motivation and Significance

1.1. The Knowledge-Extraction Bottleneck

In this paper, we present ResearchShop, a text-mining solution for ~~validated gene–disease association extraction~~ auditable gene and variant evidence extraction from biomedical literature. The tool ~~was born from a scientific necessity: genomic medicine faces a fundamental knowledge-extraction bottleneck in which~~ addresses a practical literature-triage bottleneck in genomic medicine: sequencing data are abundant, but ~~the interpretation of genetic variants relies on~~ assessing gene and variant findings often requires locating evidence scattered across a ~~literature base now measured in tens of millions of records~~ vast biomedical literature. PubMed indexes more than 40 million citations and abstracts in biomedical and life-science literature [1]. ~~Whilst curated~~ Curated resources such as ClinVar and OMIM provide essential interpretation records built from submitted, reviewed, and periodically updated evidence [2, 3]~~,~~ . ResearchShop addresses the upstream literature-triage step: newly published evidence still has to be discovered, read, and assessed before it can inform expert review or later database updates.

Połaczyłem oba zdania

This bottleneck has direct consequences. ~~In a cohort of 1,689,845 individuals undergoing hereditary disease genetic testing , 41.0% had at least one Variant~~ Large hereditary-disease testing cohorts show that Variants of Uncertain Significance (~~VUS); among reclassified VUSs, the mean elapsed time from original classification was 22.4 months for upgrades to pathogenic or likely pathogenic and 30.7 months for downgrades to benign or likely benign~~ VUSs) are common and that reclassification can lag by months to years [4]. ~~The rare-disease setting compounds this problem because~~ This problem is especially acute in rare disease, where diagnosis often depends on combining clinical observations, sequencing results, and dispersed medical literature~~; the~~ . The U.S. National Center for Advancing Translational Sciences summarizes 3–15 years as a common rare-disease diagnostic timeline, while genomic reviews emphasize that many patients remain undiagnosed even after exome sequencing [5, 6].

ResearchShop was developed from this practical need during work on paediatric inflammatory multisystem syndrome/-multisystem inflammatory syndrome in children (PIMS/MIS-C), where immune findings and genetic evidence ~~is~~ are dis-

tributed across full text, figures, tables, and supplementary material rather than concentrated in abstracts. The initial prototype was supported by a small Amazon Founders grant, but the scientific driver was operational: to bridge the gap between the scale of published genomic evidence and the practical limits of manual literature curation. The practical requirement is not only to find gene names, but to recover the surrounding evidence, provenance, and validation context needed for expert review.

1.2. Limitations of Existing Approaches

Tutaj można by trochę poprzestawiać akapity i np. informację o technikach używanych w innych narzędziach i o możliwości wykorzystania LLM-ów umieścić na wstępie, aby było wiadomo skąd te ograniczenia się wzięły

~~Already existing automated text-mining systems offer a solution but face two critical challenges~~ Automated extraction approaches range from biomedical NER and normalization tools to newer LLM-based workflows. NER systems are strong at locating biomedical entities, while LLMs can support contextual extraction and schema filling. Neither approach is sufficient on its own for auditable gene–evidence review. Two limitations motivate ResearchShop's architecture:

Full-text accessibility. Their limited abstract-only mining strategy misses information that appears in Results, Methods, tables, captions, and supplementary material. PubTator Central reported that concept taggers recovered 56–65% of manual annotations from abstracts but 78–83% from full text, an 18–22 percentage-point increase [7]. This requires robust full-text acquisition strategies.

LLM hallucinations. Large Language Models ~~show promise for~~ can support information extraction but are prone to generating plausible yet factually incorrect associations [8]. In this setting, a hallucination is not only an obviously impossible sentence. It can be a biomedical-looking identifier, such as a non-HGNC-approved "*BRCA3*" label, or a real gene symbol recalled from model pre-training but absent from the paper being processed. ~~Both failure modes are dangerous because the row may look biologically credible while lacking source-paper support~~ Pure LLM extraction can therefore introduce unsupported false positives, while overly strict validation can create false negatives by excluding real but weakly grounded findings. For biomedical review workflows, ~~therefore,~~ the relevant question is not whether the LLM can produce fluent summaries~~, but~~ . It is whether each exported association can be tied back to a validated identifier and a recoverable source passage.

Existing tools fall on a spectrum: dedicated NER and biomedical NLP systems such as PubTator Central [7], BioBERT [9], scispaCy [10], and variant-oriented tools such as tmVar 3.0 [11] support entity recognition and normalization, but they do not by themselves provide the schema-driven, auditable extraction workflow needed by researchers reviewing gene–evidence tables. LLM-based approaches offer more flexible contextual extraction, including synthesis of relationships between genes and findings, but require explicit safeguards against unsupported generations [12, 8]. ResearchShop is designed around this separation of responsibilities: NER and deterministic scanning propose source-grounded candidates, the LLM fills a user-defined schema for those candidates, and validation decides what can be exported. Table 1 compares method, full-text access, and hallucination guardrails across these approaches.

2. Software Description

2.1. System Architecture

ResearchShop is a cross-platform desktop application and a command-line utility built with Electron [13] (TypeScript/React frontend) and Python 3.11+ (extraction pipeline backend). The desktop process and Python pipeline run on the user's machine with no ResearchShop server infrastructure. The workflow ~~utilizes~~ uses direct external calls to ~~NCBI Entrez~~ PubMed through NCBI Entrez E-utilities [14], PubTator3, PubMed Central, Europe PMC, and Google Gemini as required for paper discovery, open-access retrieval, NER, and LLM extraction. Users provide their own Gemini API key and NCBI Entrez email.

The frontend communicates with the Python pipeline via an IPC (Inter-Process Communication) bridge. The pipeline process emits structured JSON messages on stdout (`PROGRESS:{...}`, `LOG:{...}`, `RESULT:{...}`) which the Electron main process parses and forwards to the React renderer. Secrets (API key, email) are passed as environment variables to the spawned Python process, never as command-line arguments.

The pipeline follows the seven implementation domains shown in Figure 1.

tę listę można by jakoś wprowadzić, bo na razie pojawia się bez wcześniejszej zapowiedzi w tekście, poza wzmianką w przedostatnim akapicie

czy dobrze rozwinąłem, co to jest Entrez?

może "full capped result set before pagination" jest zbędne?

czy to jest stała konfiguracyjna w naszym oprogramowaniu? czy można zmienić tę wartość?

1. ~~**Paper selection.** PubMed records are retrieved through NCBI Entrez E-utilities and ranked using title/abstract relevance, citation information, and journal quartiles. All records remain visible, and user-supplied PMID lists are not widened.~~
2. ~~**Open-access filtering.** Query mode considers a configurable pool of open-access candidates before selecting the requested top-*N*; selected PMIDs proceed only when full text is available.~~
3. ~~**Paper reading.** PMC JATS XML is preferred, with Europe PMC XML as fallback; sections, tables, captions, image links, and supplementary metadata are retained where available.~~

Tool or approach	Method	Full-text access	Hallucination guardrails
PubTator Central / PubTator3	Biomedical concept annotation and entity normalization	Abstracts and available full text	Strong entity normalization, but limited schema-driven contextual extraction
BioBERT, scispaCy, tmVar 3.0	Biomedical language models, entity taggers, and variant normalization tools	Depends on deployment and supplied corpus	Strong for task-specific NLP components, but no end-to-end LLM evidence audit
Unconstrained LLM extraction	Prompt an LLM to extract associations and summaries from supplied text	Possible when source text fits the model context window	Weak unless source grounding, citation checks, and identifier validation are added
ResearchShop	PubTator3, deterministic HGNC scanning, and structured LLM extraction	Open-access PMC and Europe PMC full text, including tables and figure captions where available	HGNC validation, source-text grounding, citation/table checks, confidence gating, and audit artifacts

Table 1: ResearchShop relative to common biomedical literature-extraction approaches.

4. ~~**Candidate discovery.** PubTator3 NER, deterministic HGNC symbol/alias scanning, and a full-text LLM pass propose candidates. Optional recall and figure passes can extend the set.~~
5. ~~**Detail extraction.** Gemini fills the custom schema and supplies source citations. PubTator and HGNC candidates are included as seeds. The default model is `gemini-2.5-flash-lite`, and model identifiers are configurable.~~
6. ~~**Validation.** A bundled HGNC database, HGNC REST, and MyGene.info support normalization; source grounding, citation checks, HGVS matching, and confidence gates determine exportability.~~
7. ~~**Output writing.** A bounded worker pool applies a configurable per-paper timeout and writes deduplicated CSV, JSON, XLSX, metadata, trace, and candidate-audit artifacts.~~

1. **Paper selection** retrieves PubMed records through NCBI Entrez E-utilities [14]. Topic results are ranked by gene relevance and capped impact signals; user-supplied PMID lists are unchanged.
2. **Open-access filtering** admits only papers whose full text can be fetched from PMC or Europe PMC.
3. **Paper reading** preserves JATS sections, tables, captions, image links, and supplementary metadata where available.
4. **Candidate discovery** combines PubTator3 NER [15], deterministic HGNC scanning, and a full-text LLM pass.
5. **Detail extraction** fills the user-defined schema and requests source quotations. The default model is `gemini-2.5-flash-lite`; model identifiers are configurable [16].
6. **Validation** applies a versioned bundled HGNC snapshot [17], HGNC REST [18], MyGene.info [19], source grounding, citation checks, and HGVS matching.
7. **Output writing** uses a bounded worker pool and configurable timeout, then writes primary tables and audit artifacts.

Figure 1 separates proposal, extraction, and validation because these stages have different error profiles. The UI and paper-selection stages decide *which papers* to process; the candidate-discovery stages decide *which gene labels* deserve attention; the LLM stage fills user-defined fields only after receiving source-derived candidates; and the validation layer decides *which rows* are safe enough for the primary export. Hallucinations are constrained at three points: the LLM is seeded with candidates that already came from the source text, every returned gene must be grounded back to the article, and citation-bearing fields must point to recoverable source passages before export.

2.2. Hybrid NER-LLM Architecture

~~NER identifies biomedical entities but generally does not answer the user-facing question: what did the paper claim about the gene? Pure LLM extraction can fill richer schemas, but it may attach a plausible claim to the wrong gene or to no source passage. ResearchShop therefore assigns PubTator3, HGNC scanning, the LLM, and validation separate roles. PubTator3 supplies a precision-oriented entity set; HGNC scanning adds source-grounded recall; the LLM supplies contextual fields and quotations; and deterministic checks record why each row passed or failed. Source-derived candidates are injected into the prompt, and every returned gene is searched in the article using its canonical symbol, aliases, biomedical mention bridges, and raw model label.~~

The hybrid design separates entity detection from contextual extraction. PubTator3 supplies normalized biomedical entities, while deterministic HGNC scanning recovers symbols and aliases missed by NER. The LLM fills researcher-defined fields for these source-derived candidates, but its rows remain untrusted until identifier, text, citation, and variant checks pass. Pure NER therefore lacks the requested context; pure LLM extraction lacks deterministic evidence constraints.

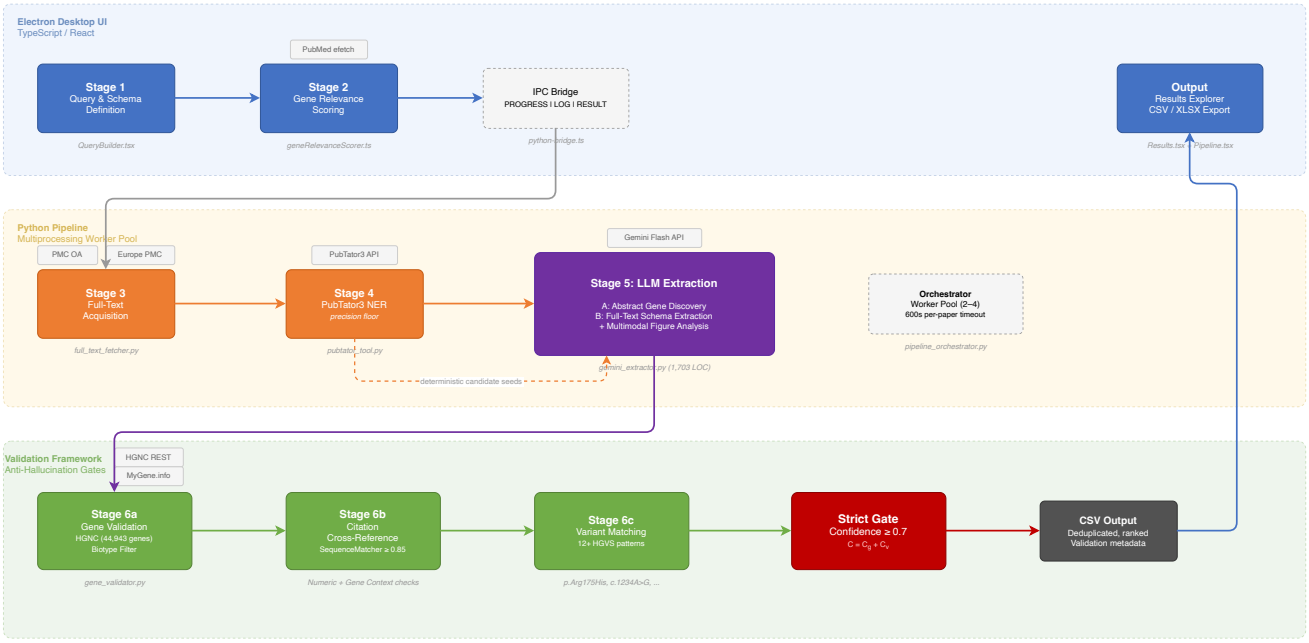


Figure 1: ResearchShop system architecture. The Electron UI layer (top) handles query input, gene relevance scoring, and results display. The Python pipeline (middle) executes open-access full-text reading, candidate discovery, LLM-based detail extraction, and output writing. The validation framework (bottom) applies gene symbol verification, source grounding, citation cross-referencing, variant checking, and a strict confidence gate ($\geq$0.7) before output.

2.3. *Schema-Driven Extraction*

Users define custom output columns as natural-language descriptions. For example, researchers might specify:

- **Key Finding**: Main research finding for this gene
- **Statistical Evidence**: p-values, odds ratios, confidence intervals
- **Functional Impact**: Effect on protein function or pathway

The pipeline dynamically constructs LLM prompts incorporating these column definitions, enabling researchers to customise extraction for different contexts (pharmacogenomics, rare diseases, cancer genomics) without code modifications. Schema-driven extraction was chosen instead of free-form summarisation because it makes the output contract explicit: every run has named fields, predictable column order, companion citation columns, and validation metadata that can be checked programmatically. For each extracted value, the LLM is required to provide a verbatim citation from the source text. If a requested field is not supported by the paper, the expected behaviour is an empty field rather than a speculative answer.

2.4. *Desktop Application*

The application is packaged with electron-builder. The current public release provides a macOS `.dmg`; Windows `.exe` and Linux `.AppImage`/`.deb` targets are supported by the build configuration and remain part of release validation. On first launch, the application automatically creates a Python virtual environment and installs pipeline dependencies.

The GUI provides:

- **Onboarding wizard:** API key validation, output directory selection, research-use disclaimer.
- **Query builder:** PubMed search with relevance scoring and paper selection.
- **Live pipeline monitor:** Real-time progress, per-stage breakdown, Gemini API usage tracking, and technical logs for debugging failed runs.
- **Results explorer:** Sortable table with confidence badges (CORROBORATED, SUPPORTED, LIMITED EVIDENCE, NEEDS REVIEW), plain-language confidence explanations, evidence/provenance fields, and CSV/XLSX export.

Security measures include OS-keychain-backed API key encryption (via Electron `safeStorage`), Content Security Policy headers, IPC path traversal validation, sandboxed renderer, and PMID input sanitisation.

Badge	Meaning
CORROBORATED	Multiple checks agree and matched supporting text was found.
SUPPORTED	Valid gene appears in the paper; some evidence may need inspection.
LIMITED EVIDENCE	Support is weak, incomplete, or available only in limited context.
NEEDS REVIEW	Important check failed, extraction was incomplete, or manual review is required.

Table 2: Confidence badges are review cues, not clinical assertions.

2.5. Validation Framework

The validation framework treats the LLM output as an untrusted draft. The model may propose structured rows and fill user-defined fields, but the pipeline decides whether a row is exportable by re-checking it against source text, controlled vocabularies, and deterministic pattern matchers. A second LLM could provide another fluent judgement, but it would not solve the core auditability problem.

2.5.1. LLM Output Validation Heuristics

For each paper, validation begins with a constrained candidate set assembled from PubTator3 entities, deterministic HGNC symbol and alias scanning, and LLM candidate discovery. Candidate generation asks which genes might be relevant; detail extraction asks what the paper says about those genes; validation asks whether each row is grounded enough to export. The returned rows are processed by the following deterministic heuristics:

1. **Identifier normalisation.** Map the reported string to an HGNC-approved symbol and retain aliases as provenance.
2. **Source presence.** Require the canonical symbol, an alias, or the raw LLM mention in the fetched text.
3. **Citation anchoring.** Match each evidence quotation back to the article.
4. **Local gene context.** Require the relevant gene near the matched quotation.
5. **Numerical preservation.** Require numeric tokens in a quotation to occur in the source passage.
6. **Variant sanity check.** Test variant-like strings against HGVS-oriented patterns.
7. **Hard gates before export.** Exclude rows that fail identifier, source, or citation requirements.
8. **Confidence label and audit trail.** Score rows that pass the hard gates and record every exclusion.

These heuristics bound the LLM's role rather than replacing expert curation: the model may summarize and structure evidence, but exported rows must remain tied to identifiers and source passages that ResearchShop can locate in the paper.

Może informację o tym, że jest to CSV można tu pominąć?

Candidate generation, schema filling, and validation are separate operations. PubTator3, HGNC scanning, and LLM discovery propose genes; the detail call supplies field values and quotations; the deterministic rules in Table 3 decide whether a row is exportable. This division limits two complementary errors: pure LLM extraction can add plausible but unsupported rows, whereas rigid entity filtering can discard genuine findings expressed through aliases or unusual context. ResearchShop therefore retains rejected candidates in audit files while admitting only source-grounded rows to the primary export.

The thresholds in Table 3 are engineering heuristics selected to favour precision in the default export, not claims about biological truth. The 0.85 citation threshold permits XML, Unicode, and line-wrapping drift while rejecting paraphrases. The ±1,500-character context window covers captions, table lead-ins, and adjacent paragraphs without accepting article-level co-mentions as evidence. The 0.7 confidence threshold is a conservative export cutoff applied only after required hard gates pass, not a calibrated probability.

Co należy rozumieć przez biologicla constants?

Gene symbol validation. Every extracted gene is checked against a bundled HGNC snapshot, HGNC REST, and MyGene.info. Fuzzy matching suggests corrections but is not authoritative. Biotype is retained as metadata, and valid non-protein-coding symbols are not rejected.

Grounding and citation checks. The canonical symbol, aliases, and raw model label are searched in the paper. Citations are checked by exact or dense sequence matching; short quotations require exact matches. Numeric consistency, local gene context, and symmetric encoding normalization are also enforced.

Variant and confidence checks. Variant strings are compared with HGVS-oriented patterns. Rows must first pass HGNC normalization, source grounding, and citation anchoring. A confidence score is then computed from gene and variant validation, and rows below 0.7 are excluded from the primary export.

Identifier validation queries the bundled HGNC snapshot first, followed by HGNC REST and MyGene.info; fuzzy matches are suggestions only [17, 18, 19, 20]. The source-grounding check then requires the canonical symbol, an accepted alias, or the raw model label in the fetched article. This blocks real genes recalled from model training but absent from the paper.

Evidence quotations are matched exactly or with normalized `SequenceMatcher` similarity of at least 0.85 [20, 21]. Quotations shorter than five words require exact matches. Numeric tokens must be preserved, and the relevant gene must occur within a ±1,500-character source window. These are conservative engineering settings, not biologically calibrated thresholds: 0.85 tolerates extraction-format drift, while the bounded window covers adjacent paragraphs and table or caption lead-ins without accepting article-wide co-mentions.

Variant strings are compared with HGVS-oriented patterns [22]. Rows must pass identifier, source, and applicable

Check	Decision rule	Failure addressed	Primary output action
HGNC normalization	Map the reported gene string to an approved symbol using the local HGNC snapshot, aliases, previous symbols, HGNC REST, or MyGene.info fallback. Fuzzy matches may suggest corrections but are not authoritative evidence.	Invalid symbols, obsolete labels, ambiguous aliases, and typographic near-misses.	Required gate; unresolved symbols are excluded from the primary ~~CSV~~ export.
Source grounding	Require the canonical symbol, an alias, or the raw LLM mention to occur in fetched article text.	Real but off-paper genes introduced from model prior knowledge.	Required gate; failures go to review/drop artifacts.
Citation anchoring	Match the LLM evidence quote to article text after normalization using exact matching or dense `SequenceMatcher` matching [20, 21].	Fabricated, paraphrased, or unlocatable citations.	Required row-level gate when citation fields are present; at least one supplied citation must ground.
Local gene context	Require the gene symbol or accepted alias within the citation context window.	Correct source sentence attached to the wrong gene in a multi-gene paper.	Enforced inside citation validation.
Numeric preservation	Require source matches for p-values, odds ratios, percentages, cohort sizes, positions, and other numeric tokens.	Altered statistics, rounded values, or invented quantitative detail.	Enforced inside citation validation when numeric claims are present.
Variant validation	Match variant-like strings against HGVS-oriented protein, coding, genomic, splice, indel, copy-number, translocation, and cytogenetic patterns.	Malformed variants or unsupported nomenclature.	Variant confidence only; unvalidated variants remain review metadata.
Confidence gate	After required gates pass, export rows only when $C_{\text{total}} \geq 0.7$.	Weakly supported associations after identifier and evidence checks.	Below-threshold rows are omitted and audited.

Table 3: Deterministic decision model used to validate LLM-generated extraction rows before export.

citation gates before scoring. For rows that pass,

$$C_{\text{total}} = C_g + C_v \tag{1}$$

where $C_g = 0.7$ for a validated gene (1.0 for gene-only rows) and $C_v = 0.3$ for a syntactically valid variant. The 0.7 export cutoff is therefore a workflow heuristic, not a probability of biological correctness. Failed rows remain in audit artifacts for review.

3. Illustrative Examples

3.1. Research Workflow

~~ResearchShop supports a compact literature-curation loop: users provide a PubMed query, author search, or PMID list; select papers; define schema columns such as *Key Finding*, *Disease Association*, *Statistical Evidence*, and *Conclusion*; and receive spreadsheet-ready gene–paper rows with provenance and validation metadata. For each selected open-access paper, the pipeline parses text, tables, captions, figures, and supplementary metadata where available; combines PubTator, HGNC scanning, and LLM candidate discovery; fills the requested schema only for grounded candidates; and exports citation status, NCBI identifiers, candidate source, association grouping, confidence label, and paper metadata. The same backend can be run through the desktop interface or from the command line for scripted analyses.~~

Users search PubMed, select open-access papers, and define the fields to extract for each gene or variant. ResearchShop then returns a spreadsheet with the requested fields, source citations, normalized identifiers, and validation metadata. The desktop interface and command-line mode use the same pipeline.

~~Figure 2 shows the protocol, query/schema, and results views. Table 4 uses a local demonstration result file to illustrate the visible data model; it is not an accuracy benchmark.~~

~~For the example in Figure 2, we searched for `KRAS AND Poland AND lung cancer` and selected a 2026 Polish cohort study of pulmonary large-cell neuroendocrine carcinoma (PMID 41977128) [23]. The run fetched the full text from PMC, extracted two tables, and produced 42 output rows representing 21 genes in 85.6 s. Two additional candidate rows were rejected because their supplied citations could not be matched to the article. Table 4 shows three retained rows; this example demonstrates the workflow, not extraction accuracy.~~

For the example in Figure 2, the query was `(KRAS AND Poland AND lung cancer) AND 2025:2026[dp]`. Table 4 reports the complete recorded run. The History view's *Genes found* value sums unique genes within each paper, whereas the Results view deduplicates gene labels across the final output. Figure 2 also shows an exported *KRAS* G12C row with separate fields and source citations. This example demonstrates output structure and traceability, not extraction accuracy. The recorded duration is retained as run provenance only. The model

and complete runtime configuration were not preserved with the supplied workbook, so this value is not compared with the controlled WSL experiment in Section 4.

3.2. Audit Artifacts

~~Each run writes the final CSV plus metadata CSV, XLSX, JSON, candidate-audit, drop-debug, and optional trace files. These artifacts record candidate origin, gate outcomes, citation-grounding status, confidence score, fallback paths, and dropped-row reasons. A reviewer can therefore see why a gene appeared, why another was excluded, and whether a supporting sentence was recovered from the paper rather than generated by the LLM.~~

To zdanie brzmi jak rasowy LLM

~~The metadata records which component proposed each row, which validation gates passed or failed, which citation matched article text, and why any row was excluded. During review, an exported row can be traced from the spreadsheet value back to its proposing source, matched citation span, normalized gene identifier, confidence label, and final gate decision without rerunning the LLM.~~

Each run includes the final tables and companion audit files. For every candidate, these files record its source, normalization, validation result, matched citation, and any exclusion reason. Reviewers can therefore check an accepted row against the article text without repeating the LLM call.

3.3. Figure and Table Handling

Many genomics papers encode primary findings in figures, tables, and supplementary files rather than in prose. ResearchShop parses PMC and Europe PMC JATS XML, extracts structured table text, collects figure captions and image URL candidates, and passes table/caption context into evidence validation. When figure analysis is enabled and images are downloadable, a multimodal Gemini model is used for analysis.

~~The parser path uses ResearchShop-owned fetchers for PubMed, PMC, Europe PMC, and citation services. A small adapter uses `pubmed_parser` for selected JATS parsing tasks such as body paragraphs and figure metadata, while ResearchShop keeps its own open-access gates, URL probing, CDN fallback, table extraction, supplementary file handling, PubTator integration, and output schema.~~ ResearchShop owns the fetchers for PubMed, PMC, Europe PMC, and citation services, as well as the open-access gates, URL probing, CDN fallback, table extraction, supplementary file handling, PubTator integration, and output schema. A small adapter uses `pubmed_parser` only for selected JATS parsing tasks such as body paragraphs and figure metadata.

Previous illustrative table and synthetic demonstration rows:

Output row	Candidate source	Displayed evidence trail	Audit interpretation
IFNG	LLM text; deterministic lexicon	Gene-normalized immune-response finding with citation status, confidence label, local HGNC validation, and matched source text.	Inspect source sentence before reuse.
HLA-C / HLA-C*04	LLM text	Allele-like input remains in the variant field while the gene label normalizes to an HGNC-compatible symbol.	Triage lead; HLA nomenclature requires expert interpretation.
CXCL10	PubTator	Sparse evidence is retained with a conservative confidence label and needs-review citation status.	Visible as a paper index, not a final assertion.

Illustrative output and audit trail from a local demonstration result file.

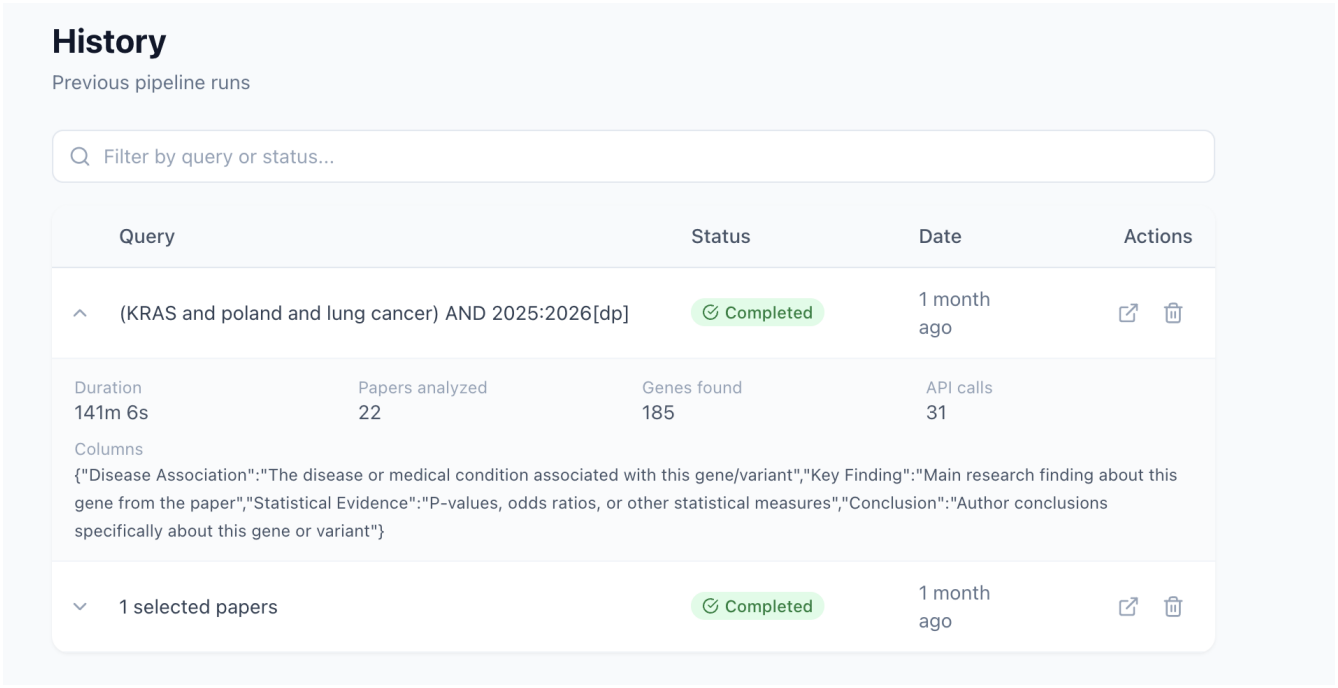
Current example generated from the documented KRAS query:

Run measure	Recorded value	Counting basis
Duration; API calls	141 min 6 s; 31	Values recorded in History for the completed run.
Papers analysed; papers in Results	22; 17	All analysed papers; papers contributing at least one final row.
Genes found; globally unique genes	185; 107	Per-paper unique counts summed; gene labels deduplicated across Results.
Candidate records; final rows	248; 216	Candidate Audit records; rows displayed in Results.
Confidence breakdown	57 corroborated; 85 medium; 2 low; 72 review 57 CORROBORATED; 85 SUPPORTED; 2 LIMITED EVIDENCE; 72 NEEDS REVIEW	Final rows mapped to the current review-cue terminology.
Citation coverage	98%	Coverage reported in the Results view.

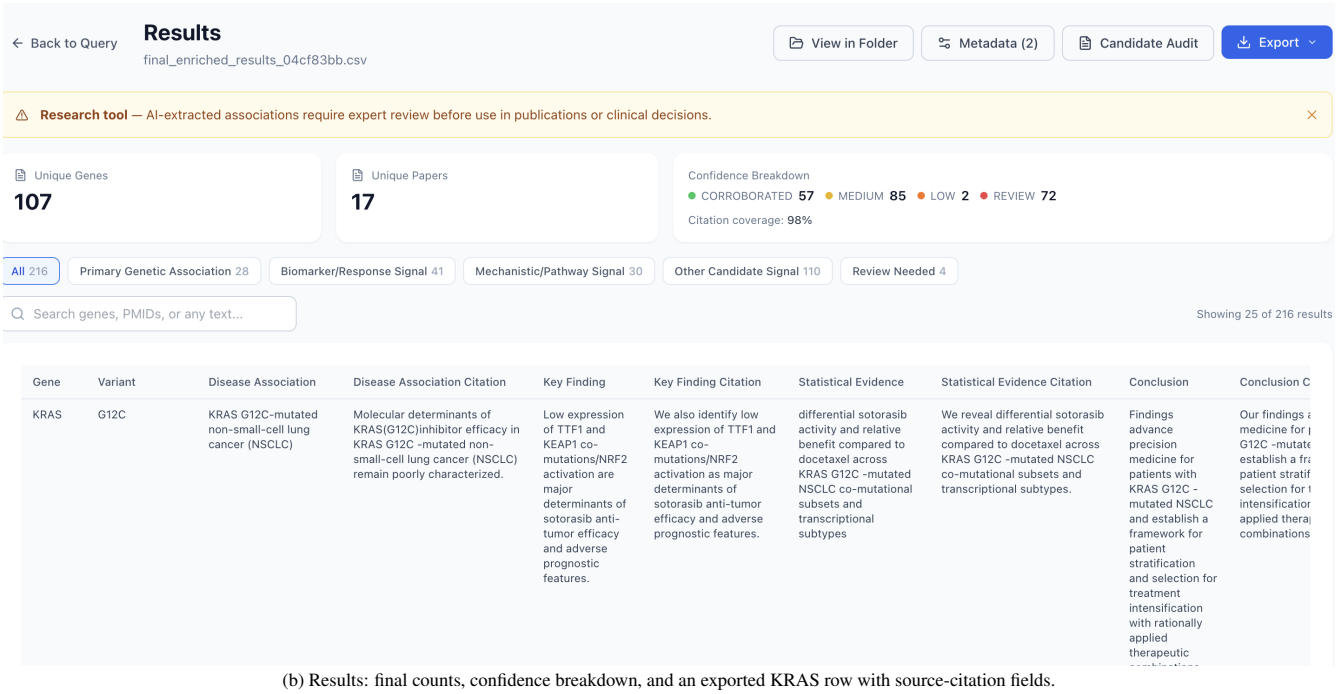
Table 4: Recorded summary of the complete KRAS query run supplied for the application example.

~~Previous three-screen workflow illustration removed.~~

Current application example:



(a) History: recorded query, duration, papers analysed, gene count, and API calls.



(b) Results: final counts, confidence breakdown, and an exported KRAS row with source-citation fields.

Figure 2: Recorded ResearchShop run for the KRAS query. Both panels use the same scale and aspect ratio so that the displayed values and output fields remain legible.

3.4. Expert Assessment of Research Usefulness

The operational checks above establish whether output can be traced to source text and whether the extraction workflow completes consistently. They do not determine whether a reported association is scientifically important or provides a useful research lead. That judgement requires review by a domain expert.

Expert assessment pending (Kasia Suski): Please select three to five representative findings from the supplied KRAS workbook, inspect their cited source passages, classify each finding as useful, not useful, or uncertain, and provide a short justification suitable for the final manuscript. No entry below has been selected or classified by the software or by AI.

Table 5: Expert assessment template for the KRAS application example. This table is intentionally incomplete in the editorial draft and must not be interpreted as an accuracy result.

Assessment item	**Expert response**
Representative KRAS findings selected for assessment	*Pending expert selection of 3–5 findings.*
Are the findings supported by the cited source passages?	*Pending expert assessment.*
Do the findings add useful information beyond the original query?	*Pending expert assessment.*
Which findings constitute useful research leads, and why?	*Pending expert assessment.*
Which findings are uninformative, misleading, or require exclusion?	*Pending expert assessment.*
Overall usefulness and suggested manuscript interpretation	*Pending expert conclusion.*

4. Impact

4.1. Primary Contribution

~~ResearchShop addresses the practical scale mismatch between biomedical literature and manual gene-evidence review. PubMed now indexes more than 40 million biomedical and life-science citations [1]; ResearchShop does not replace curation, but turns selected open-access papers into auditable gene–evidence tables that reviewers can inspect.~~

1. ~~**Hybrid extraction with auditable validation.** PubTator3, HGNC scanning, and configurable LLM extraction surface candidate genes, while grounding, citation checks, and confidence gates decide what reaches the primary export. Output rows include candidate source, normalization, validation outcome, confidence score, association type, and paper metadata; audit files preserve pre-validation candidates and gate drops.~~

Jaki jest związek tej listy z resztą tekstu? I składa się tylko z jednego punktu

ResearchShop addresses the practical scale mismatch between biomedical literature and manual gene-evidence review. Its role is not to replace curation, but to turn selected open-access papers into auditable gene–evidence tables that reviewers can inspect. The contribution is the coupling of hybrid extraction with auditable validation. PubTator3, HGNC scanning, and configurable LLM extraction surface candidate genes; grounding, citation checks, and confidence gates then decide what reaches the primary export. Output rows retain candidate source, normalization, validation outcome, confidence score, association type, and paper metadata, while audit files preserve pre-validation candidates and gate drops.

4.2. Applications

- **Variant of Uncertain Significance (VUS) evidence triage:** Helps researchers identify and review published gene-disease evidence relevant to variant interpretation.
- **Rare disease evidence triage:** Helps process curated PMID sets or focused PubMed searches where relevant reports are sparse and require source-level review.
- **Drug target discovery:** Helps researchers assemble candidate gene–disease association tables for therapeutic-development review.
- **Systematic reviews:** Produces auditable extraction tables that can support, but not replace, documented review and adjudication workflows.

4.3. Free-tier Execution Study

~~We ran the same 10 open-access PMIDs hourly for 24 Warsaw-local slots on a Windows/WSL laptop using `gemini-3.1-flash-lite`, one worker, and a three-call cap per paper. This is an operational reliability check, not a clinical accuracy benchmark.~~ We ran the same 10 open-access PMIDs hourly for 24 Warsaw-local slots on a Windows/WSL

laptop using `gemini-3.1-flash-lite`, one worker, and a three-call cap per paper. This configurable model override was the tested free-tier condition; the released application defaults to `gemini-2.5-flash-lite`. The study measures operational reliability and output stability, not biological precision or recall.

Table 6: 24-hour Gemini free-tier operational summary. Successful-batch summaries exclude the single upstream full-text failure; candidate and row dispositions use independent denominators.

Metric	Value
Hourly slots	24
Complete batches	23/24 (95.8%)
Completed papers	230/240
Median batch runtime	11.6 min
P95 batch runtime	13.8 min
Median paper runtime	0.6 min
Median calls/batch	21.0
Total Gemini calls	493
Median tokens/batch	422,304
Candidate records	8,570
Candidate records rejected	1,311
Citation-gate row drops	353
Quota-limited rows	0
Timeouts	0
Upstream failures	1
Recovered API events	95
Median gene Jaccard	0.953
Median gene-variant Jaccard	1.000

Table 7: Time-of-day comparison; runtime uses successful batches.

Block	Complete batches	Runtime min	Calls med.	API rec. runs	Upstr. fail.
Night	6/6	12.1	22.0	6	0
Morning	5/6	11.7	21.0	4	1
Afternoon	6/6	11.6	20.5	4	0
Evening	6/6	10.5	20.5	5	0

The condition was operationally usable: 23/24 batches completed, with no quota-limited rows or timeouts. Across successful runs, candidate validation retained 7,259 of 8,570 candidate records; final row decisions emitted 7,007 rows and excluded 353 at the citation gate, with no strict-gate drops. These independent operational counts do not estimate precision or recall. The logs recorded 95 recovered model-unavailable/API events; these are recovery log events, not necessarily 95 unique failed requests, so retries and explicit failure reporting remain required.

4.4. Limitations

~~The LLM used in the pipeline is non-deterministic, which limits strict reproducibility of the system. At its current stage, ResearchShop relies on PubMed-based discovery, open-access full-text availability, and the Gemini API, all of which may introduce additional biases or limitations. Rare-disease searches require particular care: the topic-search ranking uses citation and journal-quality signals that can deprioritise low-citation rare-disease papers, so users studying rare conditions should prefer direct PMID submission or carefully documented focused PubMed queries. Architecturally, however, the system remains modular and can be extended to alternative search sources or local models in future versions. Consequently, outputs generated by ResearchShop should be treated as a starting point for further investigation and always subjected to professional review and verification.~~ LLM non-determinism limits exact reproducibility, and the workflow depends on PubMed discovery, open-access full text, and an external Gemini service. Citation- and journal-based ranking may also deprioritise low-citation rare-disease reports; direct PMID lists or documented focused queries are preferable in that setting. Neither the illustrative example nor the free-tier study estimates extraction accuracy, which requires expert adjudication against a predefined reference set. Future work therefore includes an expert-labelled benchmark, alternative search sources, and local-model support for fully private processing. All outputs remain research leads requiring professional review.

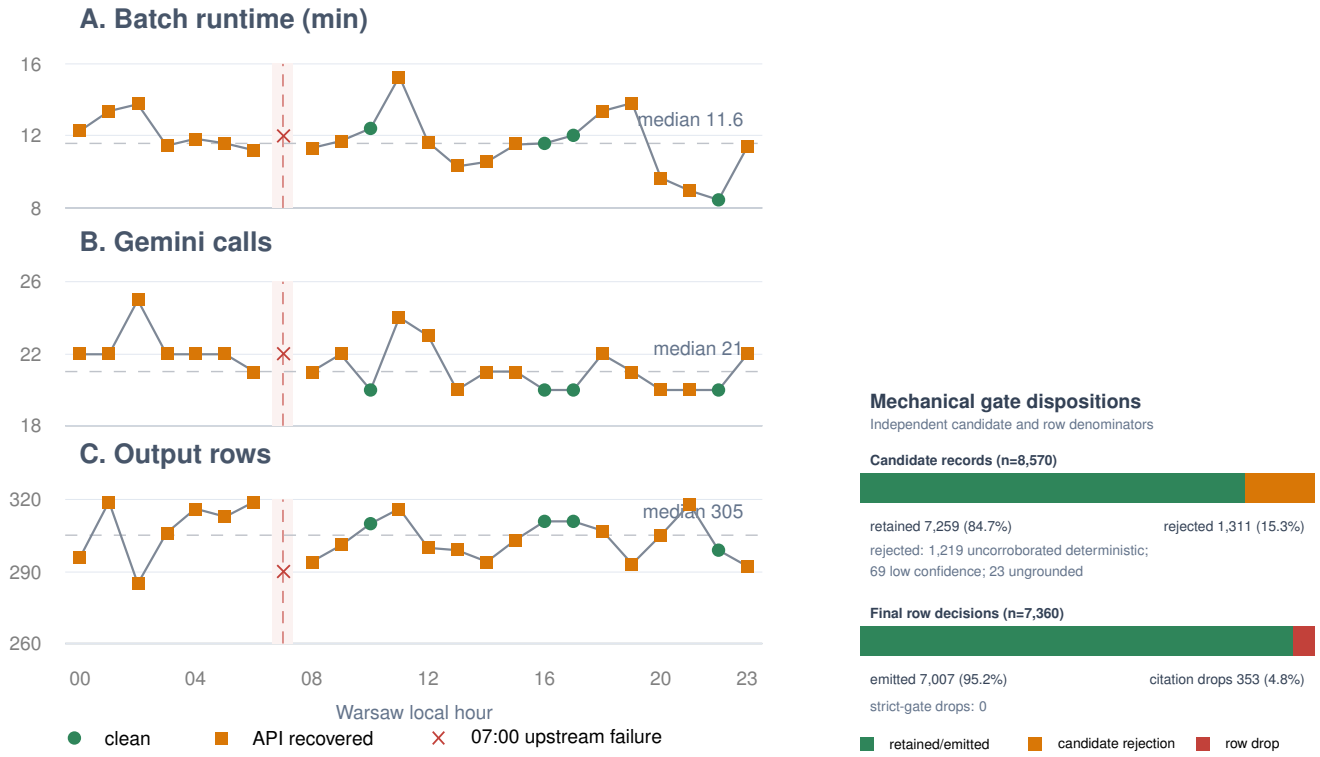


Figure 3: Gemini free-tier execution study. The hourly panel reports runtime, API calls, and emitted rows for the fixed corpus; the gate panel reports mechanical dispositions using separate candidate and row denominators. Marker color identifies clean and API-recovered runs. The 07:00 slot failed during upstream full-text acquisition, not because of quota exhaustion. Gate counts describe pipeline behavior, not biological correctness.

5. Conclusions

We have presented ResearchShop, an open-source desktop application for auditable gene and variant extraction from PubMed literature. The hybrid NER–LLM architecture combines PubTator3 candidate discovery, deterministic HGNC candidate scanning, configurable LLM-based structured extraction, open-access full-text parsing, and a multi-layer validation framework. HGNC verification, source-text grounding, variant pattern matching, citation cross-referencing, confidence scoring, and candidate audit artifacts provide defence-in-depth against common LLM extraction failures.

The application requires no ResearchShop server infrastructure; users provide their own Gemini API key and NCBI Entrez email for direct calls to external services. The project includes offline tests for parser, validation, tracing, and output behavior, plus release scripts for macOS, Windows, and Linux installers.

Future work. Planned enhancements include claim-level verification for free-text findings, LLM-assisted PubMed query construction with HGNC alias awareness, ClinicalTrials.gov integration for pharmacogenomics workflows, local or self-hosted model backends for fully private extraction when papers are supplied locally, cross-run aggregation for adjudicating uncertain rows, broader structural-variant and HLA/immunoglobulin nomenclature handling, and multi-paper relationship inference to identify consensus gene–disease associations.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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